

BT611P

Assignment 1 Fall 24

By M.Kashif(03064963933)

Total Marks: 10

Due date: 20th November 2024

Question 1:

Digital Diagram of Microbial Activity in Milk and Milk Products

Objective:

Create a detailed **digital-only diagram** illustrating the **microbial activity in milk and its products**. The diagram should cover the key types of microbes involved in dairy fermentation, spoilage, and preservation processes, as well as the roles they play at each stage.

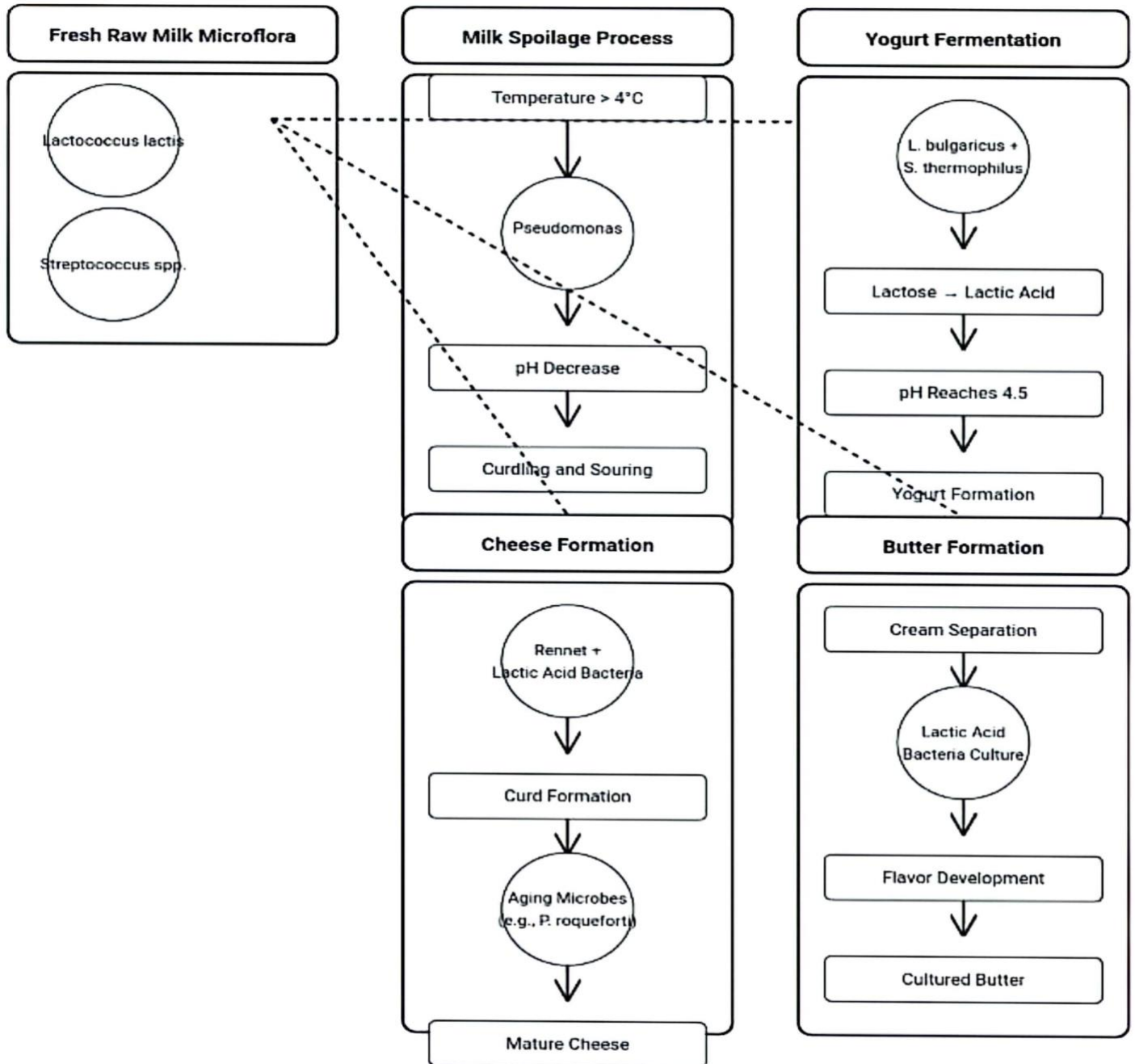
Guidelines and Requirements

1. **Software and Tools:** Use digital drawing software or tools of your choice (e.g., Adobe Illustrator, Microsoft PowerPoint, Canva, Paint or any other tools you like). Ensure that your final product is a clean, high-resolution digital illustration.
2. **Format:** Submit your final diagram and ensure the file resolution is clear and suitable for printing or display.
3. **No Text:** Submissions without any digital diagram will not be considered and will be marked zero.
4. **Original Digital Artwork Only:** All components of the diagram must be **created by you** in a digital format. Hand-drawn, AI-generated, or any internet-sourced images will not be accepted.
5. **Key Microbial Processes to Include:**
 - **Fresh Milk Microbial Flora:** Illustrate initial microbial presence in raw milk (bacteria like *Lactococcus*, *Streptococcus*, etc.) and changes that occur if milk is left untreated. (2 marks)
 - **Spoilage in Milk:** Show the microbial spoilage process in milk, indicating stages where spoilage organisms like *Pseudomonas* or lactic acid bacteria cause milk to sour or curdle. (2 marks)



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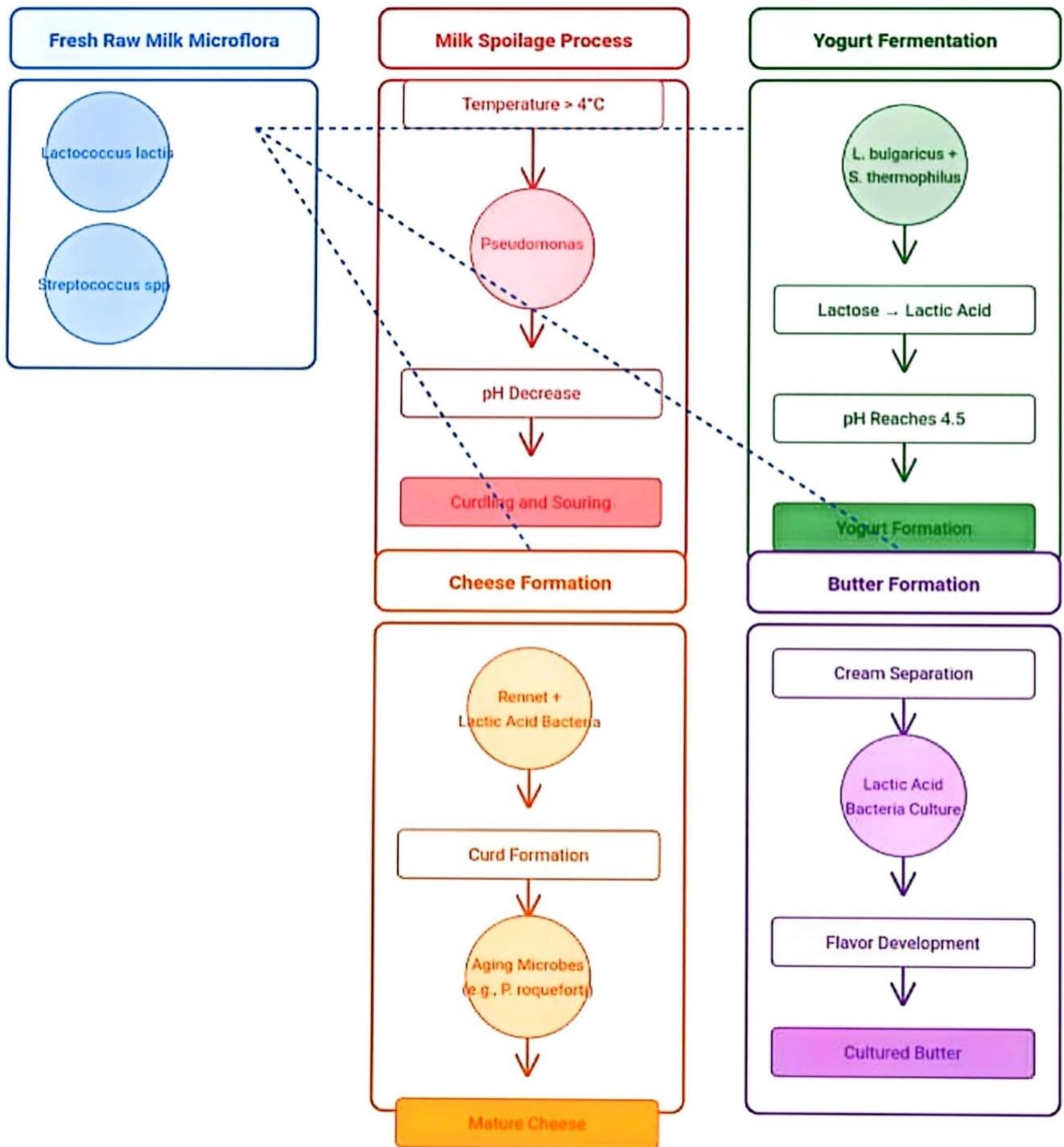
- **Fermentation for Dairy Products:**
 - **Yogurt:** Illustrate the role of *Lactobacillus* and *Streptococcus thermophilus* in converting lactose to lactic acid, giving yogurt its texture and sour taste. (2 marks)
 - **Cheese:** Depict the stages of cheese formation, including microbial roles in acidification, curd formation, and aging, with bacteria or molds as key agents. (2 marks)
 - **Butter:** Show how lactic acid bacteria contribute to the flavor and preservation in butter through fermentation. (2 marks)
- 6. **Visual Clarity:** Use distinct colors, shapes, symbols, and arrows to make each stage of microbial activity clear. Consider incorporating flow indicators to demonstrate progression in spoilage, fermentation, and product transformations.
- 7. **Creativity and Accuracy:** You are encouraged to apply creative digital design techniques to accurately represent scientific processes. Ensure each microbial activity is correctly illustrated and relevant to dairy fermentation or spoilage.
- 8. **Assessment Criteria:**
 - **Scientific Accuracy:** The diagram should accurately represent microbial roles and processes in milk and milk products.
 - **Clarity and Detail:** Each stage of microbial activity should be distinct and visually clear.
 - **Creativity:** Effective use of visual elements to convey complex processes without text.
 - **Originality:** All diagram components are original digital drawings created by you.



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